

# Krishnakumar Ramesan

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Senior Software Engineer | Distributed Systems | Multi-Agent AI | SSO & Event-Driven Architecture

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## PROFILE

Senior Software Engineer with **13+ years** building enterprise microservices, identity integrations with **SAML, OIDC, and Okta**, and event-driven systems on **Kafka and RabbitMQ**. Recently designed and shipped a production multi-agent AI onboarding system for QuickBooks on **LangGraph and OpenAI GPT 4.1**, ramped to **50% of first-time trial customers**. Strong in **Kotlin/Java**, distributed system design, data security, and production reliability at scale.

## EXPERIENCE

### Intuit — Senior Software Engineer

Aug 2022 – Present

- **Designed and shipped** a production multi-agent AI onboarding assistant for QuickBooks on LangGraph and OpenAI GPT 4.1, driving it from hackathon prototype to production as tech lead. Built a config-driven intent orchestrator that classifies each user message as an onboarding question, an expert handoff, or a general QuickBooks question beyond the domain agents' scope, then routes onboarding questions to a second config-driven orchestrator that delegates across 7 domain agents or answers directly from prior conversation and progress context.
- **Built** the screen context engine grounding those agents in what the customer sees, converting the live DOM into a compact markdown structure with on-page redaction of sensitive data like SSNs, addresses, and phone numbers. Replaced an earlier screenshot-plus-vision-LLM pipeline of roughly **9 seconds per turn with a ~30 ms DOM parse**, trading away on-screen imagery the onboarding use case did not need.
- **Hardened** the context engine for a constantly navigating app by debouncing captures to fire only after the user settles, waiting for page navigation to complete before reading the DOM, and letting agents request a fresh read on demand, keeping guidance accurate without flooding the pipeline on every interaction.
- **Designed and built** account and credential management for QuickBooks' shipping module, enabling customers to fulfill invoices end to end and print labels via a carrier aggregator. Launched to GA as the **first core revenue product in the Commerce Business Unit** with hundreds of shipping labels created post-launch.
- **Secured** thousands of carrier API credentials with AES-256-GCM envelope encryption via AWS KMS, 30-day versioned root key rotation, and mTLS in transit, built as an isolated migration-ready module to unblock GA when the platform team could not support the provider's hierarchical key model.
- **Owned** a fault-tolerant fulfillment subsystem syncing across **4 sales channels** (hundreds of thousands of events per day), designed for idempotent consumers, ordered delivery, and safe recovery under partial failures. Scaled it to sustain **75 TPS** through database partitioning, sharding, and load-based autoscaling, verifying the throughput with custom load-generation scripts.
- **Eliminated** a production 30-second timeout when bulk-matching catalog variants by profiling query patterns and removing redundant round-trips, unnecessary data loads, and missing indexes, **dropping response time from 30 s to 785 ms (97% reduction)**.
- **Designed and built** a configurable peer-review system for Live Bookkeeping expert work with FIFO-based reviewer assignment, structured questionnaires with bidirectional notifications, and UiPath RPA integration for automated final-stage review. New services onboard through configuration alone, **saving ~25,000 hours of expert time (30 min × 50K active services)**.

### TriNet — Software Engineer

Aug 2016 – Jul 2022

- **Built** a scalable ACL-based authorization service consumed by **20 microservices**, eliminating per-request database round-trips with a two-tier cache (per-user role entries invalidated on role-change events, endpoint access policies at a longer TTL), measurably reducing p90 authorization latency under production load. Delivered a companion dynamic policy dashboard allowing DevOps to update role requirements without code redeploys across all 20 dependent services.

- **Built** a core SSO microservice serving **~400K TriNet customers** across **~20 cloud software provider integrations**, resolving provider-specific SAML attributes, surfacing per-user provisioned application links, and orchestrating the final SSO handshake to enable password-free access for employees with no dedicated IT resources.
- **Eliminated** dependency on live partner environments for SSO regression testing by building a SAML integration test stub covering **~20 provider integrations**, enabling QE to run full regression suites on-demand and decoupling release cycles from third-party availability.
- **Built** an audited event-bridge connecting an on-premise HR engine to new microservices, processing **300–400K events/day** with write-ahead audit logging as a replay buffer for failure recovery. Delivered an admin dashboard for support to query, correct, and resubmit failed events, **saving 10–15 min per incident**, with a full change audit trail on every human intervention.

#### **Mazda North American Operations — Contractor**

Nov 2015 – Apr 2016

- Built RESTful microservices during a monolith modernization initiative, delivering performance improvements and feature enhancements to internal applications.

#### **Infosys Limited — Systems Engineer**

Jun 2011 – Jun 2013

- Monitored and supported production ETL jobs, migrated and tested code across lower environments, and reported daily job statistics to business stakeholders.

### **EDUCATION**

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**M.S. Computer Science**, University of Alabama in Huntsville — May 2015 | GPA **3.75 / 4.0**

**B.Tech Electronics and Communication**, Amrita School of Engineering — Jun 2011 | GPA **3.0 / 4.0**

### **SKILLS**

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**Languages:** Kotlin, Java, Python

**Frameworks:** Spring Boot, Hibernate, GraphQL (Netflix DGS)

**AI / Agents:** LangGraph, OpenAI API (GPT 4.1), multi-agent orchestration, DOM-based context engineering

**Messaging / Streaming:** Kafka, RabbitMQ

**Data:** PostgreSQL, DynamoDB, Redis

**Identity / Security:** Okta, SAML, OIDC, SSO, AWS KMS, AES-256-GCM

**Platform:** AWS, Kubernetes, Podman